SAYLANI HACKATHON - PROJECT GUIDE

HCM Lite

How the app works, how to run it,
and a check against every requirement in the brief

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Generated 25 Oct 2026 from the working project in this folder.

1. The result at a glance

HCM Lite is a web-based Human Capital Management system for three roles - **Admin, HR and Employee**. It starts with an **empty database**: the Admin sets up the company, HR registers employees, and every number on every dashboard comes from what people do in the app. It includes attendance, leave with approvals, monthly payroll with PDF payslips, three dashboards, an AI assistant that respects each role's access, announcements, notifications and an audit log.

Short answer to "have you fulfilled all requirements?"

All 25 functionalities in the brief's checklist are built and working, and all business rules in Section 3 are implemented and tested. What is **not** finished are items that need your accounts or your action, plus some optional bonus ideas:

- **Hosted (live) link** - **done**: https://hcm-lite-phi.vercel.app (Vercel + Neon). It opens at first-run setup.
- **Git repository link** - required for submission: upload the project to GitHub (see UPLOAD_TO_GITHUB.pdf in the zip).
- **Real AI key** - the AI was tested with a mock provider and the built-in fallback. Add a free Groq or Gemini key and try the live questions before the demo.
- **Optional bonus not built:** native mobile app, face matching / blink liveness, attrition-risk score, absence-anomaly flags.

Area	What was checked	Status
25 functionalities (Section 8)	Every row built with the exact Admin / HR / Employee access in the checklist	Done
Business rules (Section 3)	M1-M9 rules, including 409 / 422 codes, quotas, payroll formulas and locking	Done
AI assistant (Section 4)	Chat with 7 tools, dashboard summary, smart leave, policy Q&A, guardrails	Done
AI with a real provider	Only mock + built-in fallback tested (no key was available)	Your step
Metrics & dashboards (Section 7)	All formulas, filters by date and department, server-side	Done
Camera + location attendance	Web version: face count + GPS + server distance check	Done
Mobile app (APK / Expo)	Optional bonus	Optional
Hosted link	https://hcm-lite-phi.vercel.app - live, opens at first-run setup	Done
Git repository link	Upload to GitHub (UPLOAD_TO_GITHUB.pdf)	Your step
README a judge can follow	README.md in the project	Done
Tests	128 unit + integration tests, 8 end-to-end browser tests - all passing	Done

Verification that was actually run

- Type check, lint and production build: **pass**.
- 128 automated unit + integration tests against a real PostgreSQL database: **pass**.
- 8 Playwright browser tests that perform the brief's whole 5-minute demo through the real UI: **pass**.
- A 57-step API test of every role boundary (for example, an employee changing an ID in the URL gets 403): **pass**.
- Screens checked at desktop width and at 375 px phone width.

2. Run it and log in

What you need: **Node.js 20.9+** and **PostgreSQL** (already on this computer, port 5432).

Start the app

```
Open a terminal in the project folder and run:
npm install
npm run db:migrate
npm run dev
```

Then open **http://localhost:3000** in a browser. The **.env** file is already configured for this computer.

Login credentials - there are none by default

The brief says the system must start empty, so **there is no built-in username or password**. You create them:

Step	Who	What happens
1	You	First visit shows **First-run setup**. Enter a company name, an Admin email and a password (8+ characters, a letter and a number). You are now signed in as **Admin**.
2	Admin	**Organization** page: add at least one department and one designation (and office time / location if you want).
3	Admin	**User accounts -> Create HR login**. A **temporary password** is shown once - copy it.
4	HR	Sign in with that email + temporary password. The app forces a new password.
5	HR	**Employees -> Register employee**. Each employee also gets a temporary password shown once.
6	Employee	Signs in, sets a new password, then can check in, apply for leave, download payslips.

Ready-made test accounts (optional)

The last automated browser test left a **separate test database** (hcm_lite_e2e) filled with sample activity. To look around without creating anything, start the app against it:

```
$env:DATABASE_URL="postgresql://postgres@localhost:5432/hcm_lite_e2e?schema=public"; npm run dev
```

Role	Email	Password
Admin	admin@demo.test	AdminPass1
HR	hr@demo.test	HrPassword1
Employee	ali@demo.test	AliPassword1

This test database is wiped and rebuilt each time **npm run test:e2e** runs, so treat these as throw-away. **Forgot a password?** The Admin opens **User accounts** and clicks **Reset**; a new temporary password is shown once.

Turning on the real AI

Without a key the app still works using a clearly labelled *Built-in assistant*. For real AI, put a key in **.env** and restart. Free option (Groq) below; Google Gemini or Anthropic Claude (AI_PROVIDER=anthropic, AI_MODEL=claude-opus-5-5) also work. The key stays on the server and is never sent to the browser.

```
AI_PROVIDER=openai
AI_BASE_URL=https://api.groq.com/openai/v1
AI_MODEL=llama-3.3-70b-versatile
AI_API_KEY=gsk_...your key...
```

3. How the app works

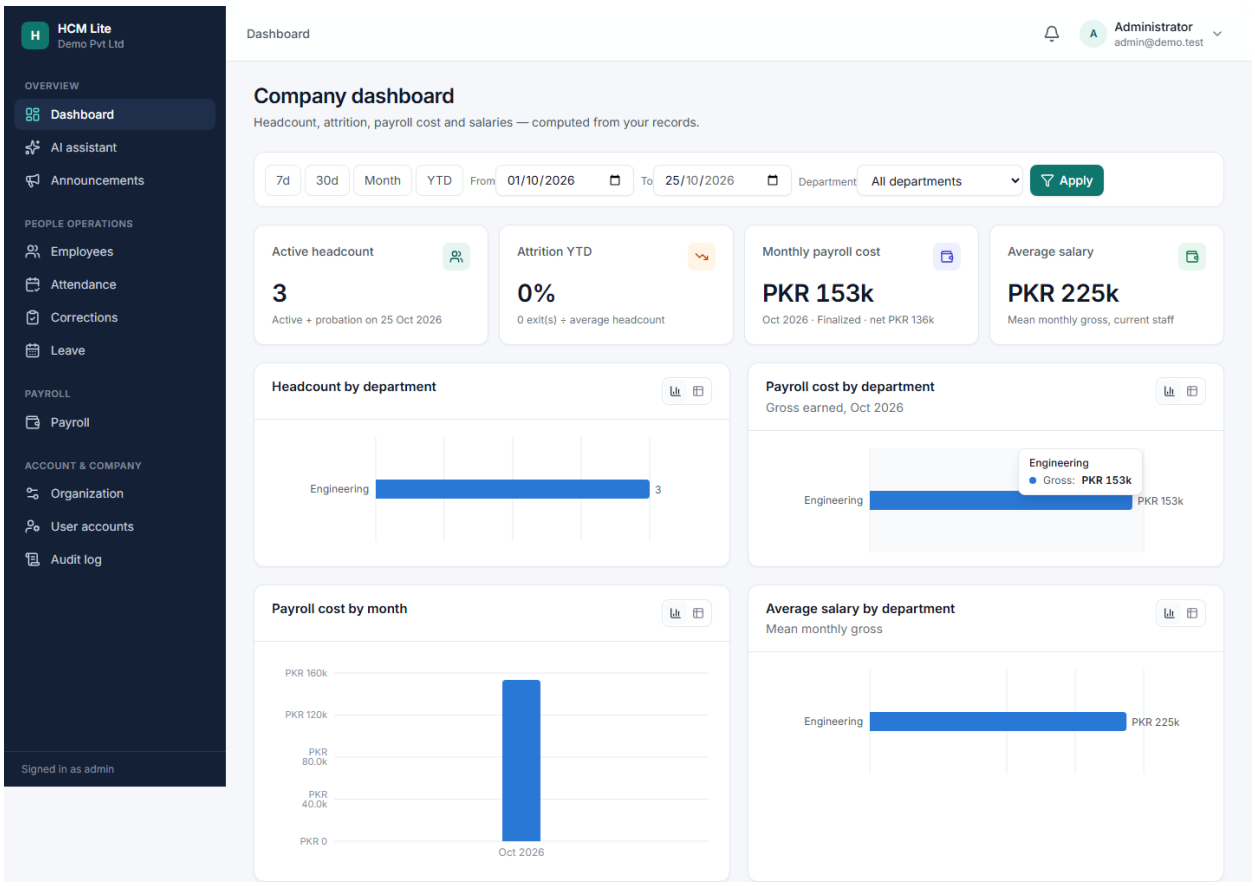
The three roles

Role	Main job	Can NOT
Admin - company level	Company setup (office hours, grace, weekend, office location), departments, designations, leave types, holidays; creates HR logins; deactivates / resets any login; finalizes payroll; company dashboard; audit log; announcements; AI on all data.	Edit a payroll run after it is finalized; change or delete audit entries.
HR - people operations	Registers employees (creates their login), edits details, salary structures, CSV import / export; daily attendance board; approves leave and attendance corrections; prepares the payroll draft; HR dashboard; announcements; AI on HR data.	Change company settings, create Admin users, finalize payroll, view the audit log.
Employee - self-service	Check in / out (web, or camera + location); own attendance calendar; request corrections; apply for / cancel leave, smart leave; download own payslips; edit own phone and address; personal dashboard; AI on own data and policy.	See anyone else's data or salary; approve anything; change own salary, role or attendance directly.

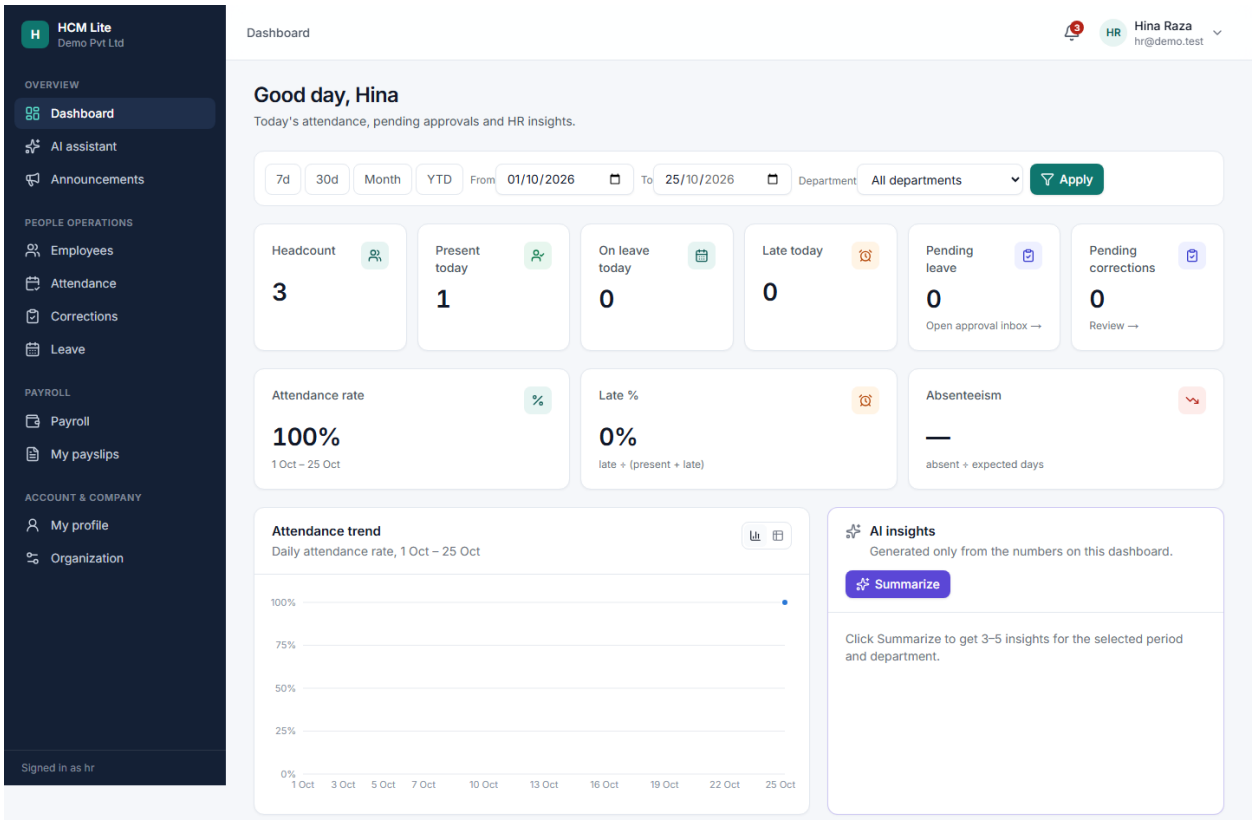
HR staff are also employees (they check in and take leave), so an HR login comes with an employee profile. HR cannot approve their own leave or change their own salary. Every one of these limits is checked on the **server**, not just hidden in the menu.

The demo flow, step by step

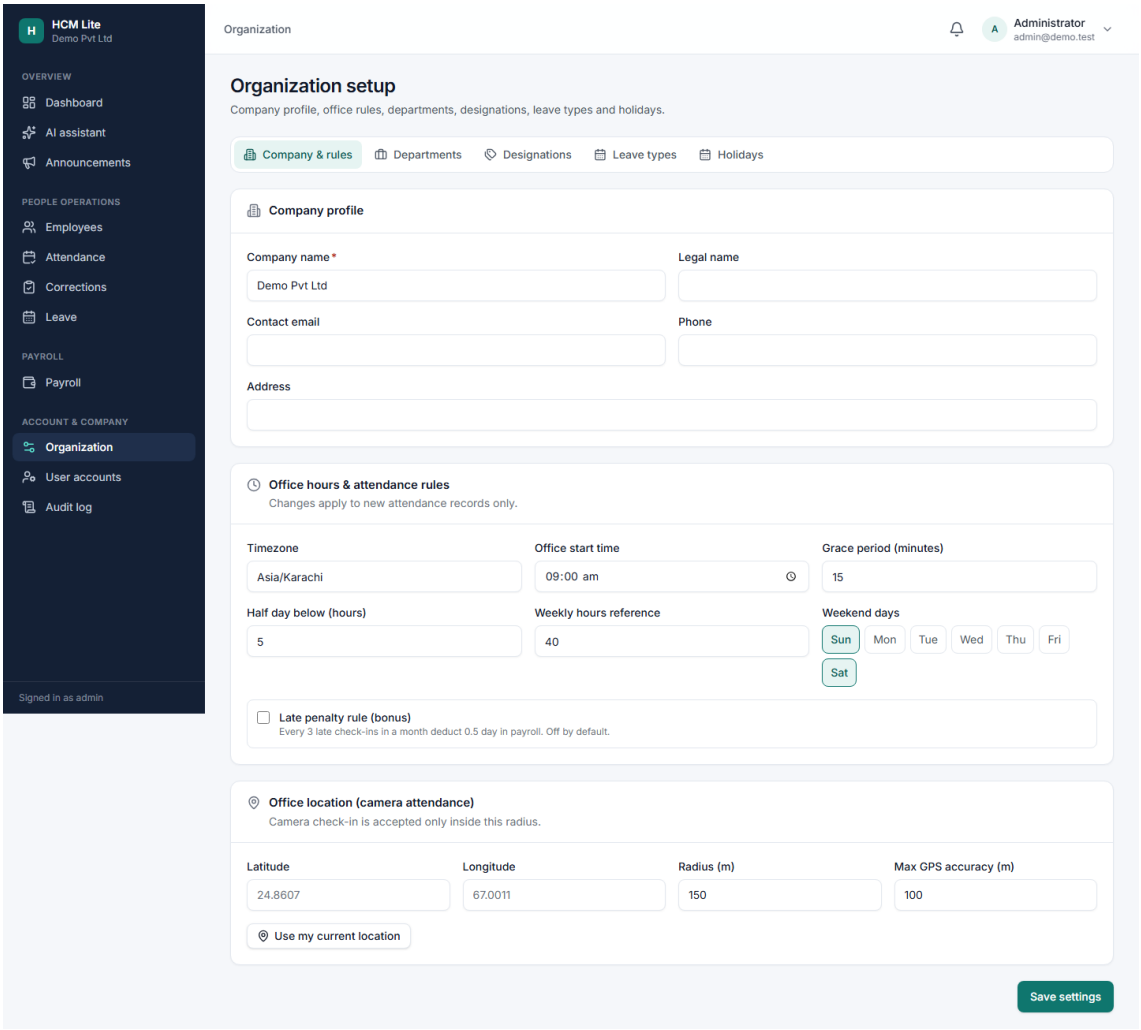
#	Who	Where in the app	What to do
1	Admin	First-run setup -> Organization	Create company; add a department + designation; set office hours / location; review leave types.
2	Admin	User accounts	Create HR login; note the temporary password.
3	HR	Employees -> Register employee	Register 2-3 employees (each gets a login).
4	Employee	Dashboard	Sign in, change password, click **Check in**.
5	Employee	Leave -> Smart leave	Type "I need next Monday and Tuesday off for my sister's wedding" -> **Fill the form** -> check -> **Confirm & submit**.
6	HR	Leave -> Approval inbox	**Approve** with a comment (employee is notified, balance goes down).
7	HR	Dashboard / AI assistant	Click **Summarize**; ask "Who is on leave today?".
8	HR	Payroll	**Prepare draft** for the month; review the employee-wise table.
9	Admin	Payroll -> open the month	**Finalize payroll** (locks it, publishes payslips).
10	Employee	My payslips / AI assistant	Download the **PDF**; ask for a colleague's salary -> polite refusal.
11	Admin	Audit log	Show who did what, with before / after data.



Admin company dashboard - headcount, attrition, payroll cost, average salary, charts and audit feed.

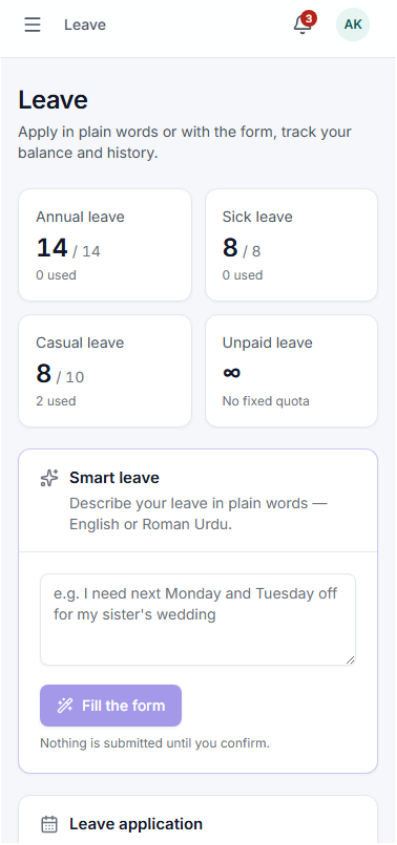


HR dashboard - today's numbers, pending approvals, attendance rate, trend chart and AI insights.

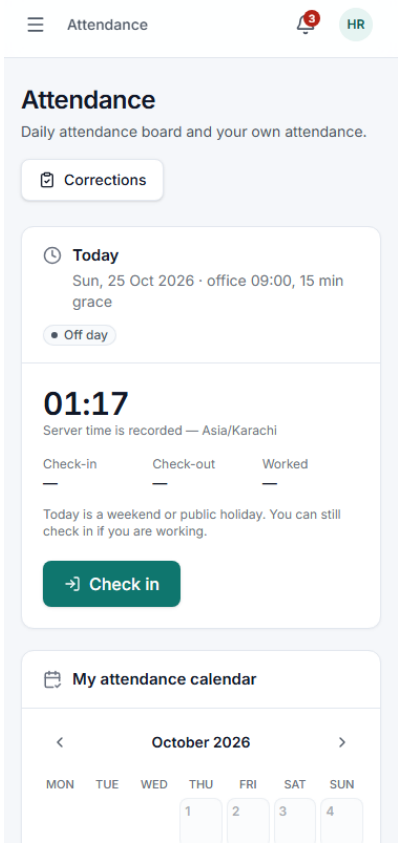


Organization setup (Admin) - office hours, grace period, weekend, office location and radius.

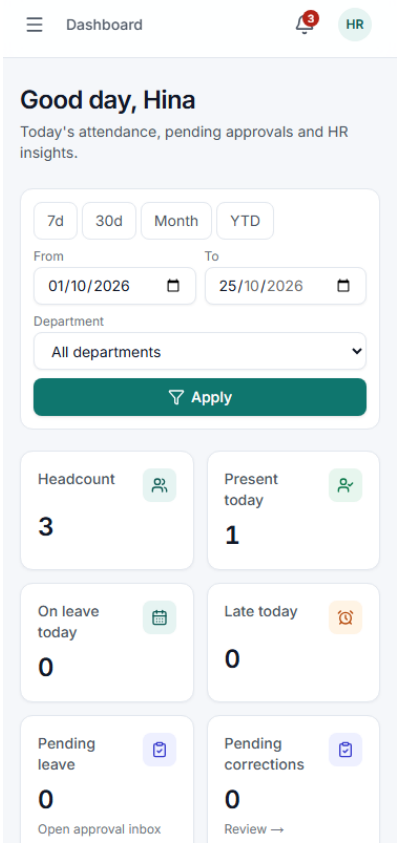
Works on phones (375 px and up)



Employee leave page on a phone



Check-in widget on a phone



HR dashboard on a phone

4. The 25 functionalities (brief Section 8)

Access shown as in the brief: **Manage** = create and change, **All** = every record, **Own** = only their own, **-** = no access. "Where" is the page in the app.

#	Functionality	Admin / HR / Emp	Where / how	Status
01	First-run setup: company profile + Admin account	Manage / - / -	First visit -> /setup (or ADMIN_EMAIL / ADMIN_PASSWORD env). Second setup is blocked (409).	Done
02	Login, logout, change password; role-based menus and dashboards	Own / Own / Own	Login page, account menu, Change password; each role has its own menu and dashboard.	Done
03	Company settings: office hours, grace, weekend, office location + radius	Manage / View / -	Organization -> Company & rules (HR sees it read-only).	Done
04	Departments, designations, leave types + quotas, public holidays	Manage / View / -	Organization tabs. Default quotas 14 / 8 / 10 / unpaid unlimited.	Done
05	Create HR logins; deactivate or reset any login	Manage / - / -	User accounts. Deactivation signs the user out immediately.	Done
06	Register an employee (creates login) and edit details	View / Manage / own phone + address	Employees -> Register; Employee 360 -> Edit; My profile for phone/address.	Done
07	Employee directory with search + filters; Employee 360	All / All / Own	Employees (server-side search, filters, paging); click a name for the 360 view.	Done
08	CSV import with dry-run error report; CSV export	- / Manage / -	Employees -> Import CSV (row + field + reason report) and Export CSV.	Done
09	Check in and check out on the web	- / Own / Own	Dashboard and Attendance page widget.	Done
10	Attendance calendar and daily attendance board	All / All / Own	Attendance page: own calendar; daily board for HR / Admin.	Done
11	Attendance correction: request and approve	View / Approve / Request	Corrections page. Approval is audited with before / after.	Done
12	Apply for or cancel leave; balance and history	- / Own / Own	Leave page.	Done
13	Approve or reject leave with a comment	View / Approve / -	Leave -> Approval inbox.	Done
14	Salary structure per employee	View / Manage / -	Employee 360 -> Salary structure.	Done
15	Monthly payroll: create and review the draft	View / Manage / -	Payroll -> Prepare draft; run page shows every employee.	Done
16	Finalize payroll (locks the month)	Manage / - / -	Payroll run -> Finalize payroll. Re-finalize = 409, edit after lock = 423.	Done
17	Payslips with PDF download	All / All / Own	My payslips; Employee 360; payroll run page.	Done
18	Dashboards, KPIs, trends, insight lists with filters	Company / HR / Personal	Dashboard (date range + department filters).	Done
19	AI assistant chat answering from real data	All data / HR data / Own data	AI assistant page; employee dashboard panel.	Done
20	AI dashboard summary	Yes / Yes / -	Dashboard -> Summarize.	Done
21	Smart leave: sentence -> pre-filled leave form	- / Own / Own	Leave -> Smart leave -> Fill the form -> Confirm.	Done
22	Policy Q&A on attendance, leave and payroll rules	Yes / Yes / Yes	AI assistant (answers only from the company's rules).	Done
23	Announcements	Post / Post / Read	Announcements page + on all dashboards.	Done
24	In-app notifications (leave decisions, payslips)	Own / Own / Own	Bell icon -> Notifications.	Done
25	Audit log of create, update, approve, finalize	View / - / -	Audit log (cannot be edited or deleted, even in the database).	Done

5. Requirements check: rules, AI, metrics and bonus

Business rules (brief Section 3)

Module	Rule in the brief	How it works in the app	Status
M1 Auth	JWT login, bcrypt passwords; every API checks role AND data scope	Signed cookie session re-checked against the database on every request; employees only read rows with their own employee id (tested).	Done
M2 Setup	Office 09:00, grace 15 min, half day under 5 h, weekend Sat-Sun, office location + radius; settings apply to new records only	All editable by Admin; each attendance record stores the rules used when it was created.	Done
M3 Employees	Registering creates a login with a temporary password; server-side search, filters, paging; CSV dry run checks required fields, unique code / email, valid email, known dept, ISO date, salary > 0	All checks implemented; each bad row reported with row number, field and reason; nothing is written until the file is clean.	Done
M4 Attendance	Server stamps time; one record per day (second check-in = 409); late after 09:15; half day under 5 h; absent; missing checkout	Implemented exactly; 09:15 is on time, 09:16 is late.	Done
M5 Leave	Working days only; quotas 14 / 8 / 10 / unpaid unlimited; overlap or over-balance = 422	Implemented; two approvals at the same moment cannot overspend a balance (tested).	Done
M6 Payroll	Gross = 1.5 x basic; per day = gross / working days; deduct absent + unpaid + 0.5 x half days; prorate joiners / leavers; tax slabs on gross x 12; Admin finalizes; locked	Implemented in whole paisa (no rounding drift); every tax slab tested.	Done
M7 Dashboards	All numbers computed on the server; filter by date range and department	Yes - the AI uses the same functions, so answers match the dashboard.	Done
M9 Comms & audit	Announcements on all dashboards; notifications; audit log with actor, action, entity, before / after JSON	Done; the database itself rejects edits / deletes of audit rows.	Done

AI assistant (brief Section 4)

Requirement	In the app	Status
Chat -> POST /ai/chat, model picks a tool, backend runs it with the user's scope	POST /api/ai/chat with the 7 tools: get_headcount, get_attendance_summary, get_late_comers, get_leave_summary, get_my_leave_balance, get_payroll_summary, search_employees	Done
Dashboard summary: 3-5 bullets from the KPI JSON only	Summarize button; answer validated; empty data gives a clear message	Done
Smart leave -> JSON {type, start, end, reason} -> pre-filled form, user confirms	Validated JSON, retry once, then fallback parser, then normal form; never auto-submits	Done
Policy Q&A only from the rules, otherwise "I don't know"	Rules are generated from the company's live settings	Done
API key only on the server (AI_API_KEY)	Never sent to the browser	Done
Model never queries the database; tools reuse the same permission checks	Tools call the same services as the REST API	Done
15 s timeout, friendly fallback, ~20 requests / user / hour, ai_logs, 'AI-generated' label	All implemented (prompt text is not stored in logs)	Done
Live judge questions (headcount by dept, late-comers, attendance rate, sick leaves left, colleague's salary refused, smart leave)	All answered correctly in automated tests using the built-in assistant and a mock model	Done

Requirement	In the app	Status
Tested with a real hosted model	No key was available during the build	Your step

Metrics (brief Section 7)

Metric	Formula used (same as the brief)
Headcount / attrition	Active + probation on the end date / exits in period divided by average headcount x 100
Attendance rate	(present + late + missing_checkout + 0.5 x half_day) / (working-day records - on_leave)
Late % / absenteeism	late / (present + late) / absent days / expected working days, per department
Leave utilization	Approved days / quota, per leave type and department
Payroll cost / average salary	Sum of gross and net by month and department / mean gross by department and designation

If a formula would divide by zero (for example no records yet), the dashboard shows "-" instead of a fake 0% or 100%.

Optional mobile / camera and bonus points (brief Sections 5 and 8)

Bonus item	In the app	Status
Camera attendance with face detection + location	Web (laptop or phone browser): exactly one face, GPS accuracy <= 100 m, server Haversine distance <= radius; no photo stored; users told what is captured	Done
Mobile app (APK / Expo)	Not built - the web app works on phones instead	Optional
Face matching to enrolled face + blink liveness (extra bonus)	Not built	Optional
Late penalty rule (3 lates = 0.5 day)	Organization -> switch on; used in payroll	Done
AI understands Roman Urdu	System prompt + built-in parser (kal, parson, bukhar, chutti...)	Done
Dark mode / PWA	Dark mode in the account menu; web app manifest (no offline mode)	Done
Attrition risk score	Not built	Optional
Absence anomaly flags	Not built	Optional

6. Submission and judging criteria (brief Section 9)

Brief asks for	Status now	Status
Web app with three roles, own menu, data scope and dashboard	Built	Done
README with stack, how to run, first Admin, AI key, done / not done	README.md in the project folder	Done
Commit .env.example only, never real keys	.env is in .gitignore; .env.example has placeholders	Done
Git repository link (public or shared with judges)	Upload with GitHub Desktop (UPLOAD_TO_GITHUB.pdf)	Your step
Hosted (live) link that opens at first-run setup	https://hcm-lite-phi.vercel.app	Done
5-minute live demo	Script in Section 3 of this guide	Your step
Mobile app APK / Expo link	Optional - not built	Optional

Hosting (already done)

- Live app: **https://hcm-lite-phi.vercel.app** - hosted on Vercel, database on Neon (both free).
- It opens at **Set up your company**, exactly as the judges expect. Do not complete setup there before judging.
- Code changes are not picked up automatically from GitHub; they need a redeploy from the computer the site was deployed from.

Scoring criteria and where the project stands

Criteria (points)	What judges look at	Where it is covered
Core HR flows (25)	Login per role, employees, check-in, leave approval, payroll end to end	All built; the full flow is an automated browser test
Data & analytics (20)	Dashboards match records; filters work; useful insights	Server-side formulas, date + department filters, table view for every chart
AI features (15)	Correct live answers, role scoping, smart leave, summaries, graceful failure	Built and tested; **add a real key and rehearse**
Code quality (15)	Module structure, rules in services, validation, tests	Rules in src/domain, services in src/server, Zod everywhere, 136 tests
UI / UX (10)	Clarity, consistency, loading / empty / error states, responsive	Design system, skeletons, empty states, 375 px checked
Security & RBAC (10)	Server-side scope checks, hashed passwords, AI key on server, audit log	All present; ID-tampering tested; append-only audit enforced by the database
Demo & docs (5)	Clear story; working hosted link; README	README (with live link) + this guide

7. What is not done yet / your action items

Before you submit (in this order)

#	Action	Why	Time
1	Get a free AI key (Groq or Gemini), put it in .env, restart, and ask the live judge questions	AI was never run against a real model	10 min
2	Upload the project to GitHub (UPLOAD_TO_GITHUB.pdf)	Repository link is required	10 min
3	Hosted link is already live: https://hcm-lite-phi.vercel.app	Do NOT complete setup there before judging	-
4	Rehearse the demo locally (or on the live link, then ask for it to be reset)	Judges start from a fresh install	10 min
5	Submit both links on lms.saylaniit.com	Submission	2 min

Honest limitations (also listed in the README)

- **No native mobile app** (optional). Camera check-in works in phone browsers instead.
- **Face detection runs in the browser**, so a modified browser could lie about the face count. The server still checks GPS accuracy, distance and time. Face matching and blink liveness are not built.
- **No emails**: temporary passwords are shown once on screen to the Admin / HR who created the account.
- Login brute-force protection is kept in server memory (resets when the server restarts); the AI rate limit is stored in the database.
- If a salary changes in the middle of a month, payroll uses the salary that applies at month end.
- Attrition-risk score and absence-anomaly flags (bonus ideas) are not built.

8. Design decisions, architecture and tests

Decisions where the brief left room (all documented)

- **Company timezone** (default Asia/Karachi) defines "today", late marks and smart-leave dates.
- **Absences count from the go-live day** (the day the company was set up in the app). Otherwise an employee registered with an older joining date would look absent for every earlier day and lose that pay.
- **Today is not over**: a missing check-in only becomes "absent" the next day; until then it shows "Not checked in".
- **Weekend / holiday check-ins** are allowed and count as attendance, but never create absences.
- **Leave** cannot cross two calendar years, can be backdated at most 30 days, and approved leave can be cancelled only before it starts (days return to the balance).
- **Your prompt vs the PDF**: where they differed, the PDF was followed - Admin only *views* leave approvals, corrections, salaries and payroll drafts; CSV import/export is HR-only; Admin has no check-in.

How the code is organised

Folder	What is in it
src/domain	Pure business rules (attendance, leave, payroll, tax, metrics, permissions, smart leave, CSV checks) - no database, fully unit tested
src/server/services	One module per feature; every function receives the signed-in user and checks permission + data scope
src/server/ai	AI provider adapter (Anthropic or OpenAI-compatible), the 7 tools, prompts / policy, built-in fallback
src/app/api	Thin REST endpoints that call the services
src/app/(app)	Pages for each role
src/components	Design system: cards, KPI tiles, tables, badges, dialogs, charts, chat panel
prisma	Database schema and migrations (including the audit-log protection trigger)
tests	Unit, integration (real database) and end-to-end (browser) tests

Commands

```
npm run dev # start for development
npm run build; npm start # production
npm run typecheck # TypeScript check
npm run lint # code style
npm test # 128 unit + integration tests (uses hcm_lite_test)
npm run test:e2e # 8 browser tests (needs npm run build; uses hcm_lite_e2e)
```

Databases on this computer

Database	Used for
hcm_lite	The real app - currently empty so it opens first-run setup
hcm_lite_test	Automated unit / integration tests (wiped by the tests)
hcm_lite_e2e	Automated browser tests (wiped and refilled by npm run test:e2e)